Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY
And
KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Rebuttal Testimony

of

Gas Load Forecasting Panel

Date: September 22, 2023

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I. Introduction

Q. Please identify the members of the Gas Load Forecasting Panel.

A. The Panel consists of Theodore Poe, Jr, Shilpa Sethia, and Thomas Gumbley.

Q. Is this the same Panel that previously submitted direct testimony and corrections and updates testimony in these proceedings?

A. Yes. The defined terms in the Panel's direct testimony and corrections and updates testimony have the same meanings in this rebuttal testimony.

II. Purpose of Testimony

Q. What is the purpose of your rebuttal testimony?

A. The purpose of our rebuttal testimony is to respond to certain recommendations in the testimony of the Department of Public Service Staff ("Staff") Forecasting and Inflation Panel ("SFIP") and certain recommendations in the testimony of Mark D. Kleinginna on behalf of Alliance For A Green Economy ("AGREE"). Specifically, we will address the normal year and design standards challenged by both parties, as well as Mr. Kleinginna's comments regarding the forecast growth in design day sendout. We will also address the SFIP's proposed forecasts and explain why the Companies' standards and forecasts should be adopted for ratemaking purposes in these proceedings.

Q. Are you sponsoring any exhibits as part of your rebuttal testimony?

A. No.

III. Companies' Response

Q. Regarding the Companies' normal year standard for heating degree days ("HDD"), do you agree with the SFIP's proposal (at 14-15) to redefine normal from a 30-year average to a 10-year average?

A. No, the Companies' do not agree with the SFIP's proposal to use a 10-year average for HDD. The Companies' forecasts were based on the 30-year average of daily HDD data for calendar years 1992 through 2021. This approach is consistent with prior National Grid rate cases, where the rate plans approved by the Commission relied on demand forecasts using the 30-year normal HDD definition for both the Companies and National Grid's upstate New York affiliate, Niagara Mohawk Power Corporation d/b/a National Grid ("NMPC").

Instead, the SFIP's proposed gas load forecast is based, in part, on the assumed definition of a normal annual HDD of 4,438, as calculated using the Companies response to Information Request ("IR") No. DPS-263. This value was determined using a 10-year average of the Companies' daily HDD data for calendar years 2012 through 2021.

Q. Why does the SFIP recommend using a 10-year weather normalization instead of the Companies' proposed 30-year weather normalization?

A. The SFIP (at 14-15) states:

> Compared to a 30-year average weather normalization, a 10-year average weather normalization more accurately reflects weather trends and is more likely to accurately predict the weather in the forecasting period. A 30-year average incorporates older data that becomes less relevant when changing climate patterns are taken into consideration, making the 30-year average less representative over time. Given the trend of increasing temperatures, using a 30-year weather average could improperly overstate the anticipated number of heating degree days in the Rate Year forecast period, resulting in a less accurate gas delivery forecast.

Q. What is the Companies' response to the SFIP's argument?

A. The Companies disagree. The Companies believe the normal year HDD standard should be a stable reference that does not change significantly over time. With the 30-year HDD, variations in pricing of gas service will be driven more by the change in the underlying costs and less by the year-to-year fluctuations in weather. Using only 10 years of data to define a normal year reflects interannual weather variability instead of long-term climate trends. A 10-year moving average will show both decreases and increases in normal HDDs over time (*See* Figure 2, below); whereas 30 years of data better reflects climate trends. Figure 1, below, illustrates how a 30-year moving average shows a consistently decreasing trend with less variability than the 10-year moving average. For this reason, the National Oceanic and Atmosphere Administration ("NOAA") defines climate

normals using a 30-year average, updated once every 10 years.[1] Using the Companies' database of daily HDD data from 1967 through the present, Figure 1, below, shows the calendar year total HDD for the Central Park, New York weather station ("KNYC") and its 30-year normal. The 30-year average in Figure 1 indicates a slow descent from 1997 into the mid-2010s and a slowing in more recent years.

Figure 1

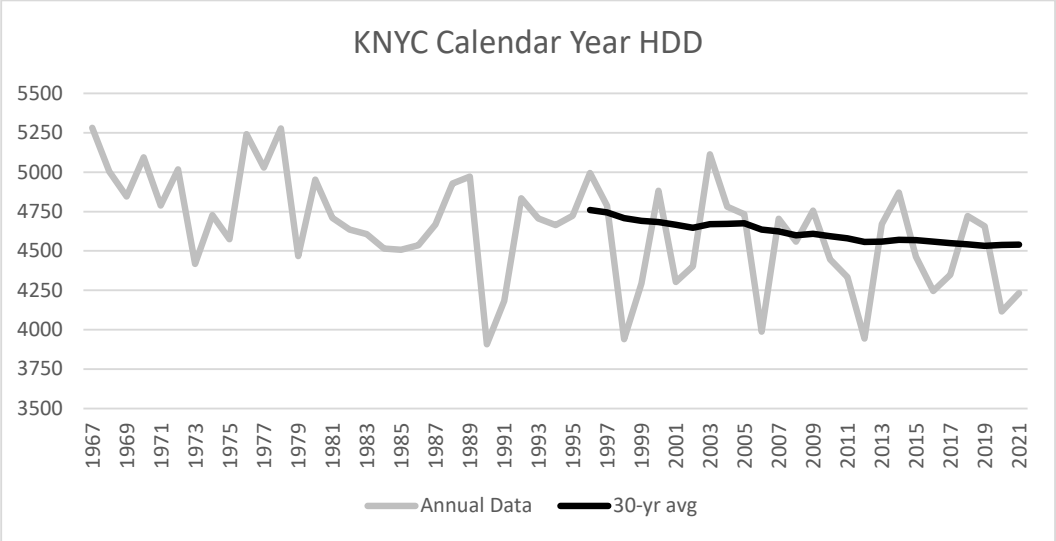
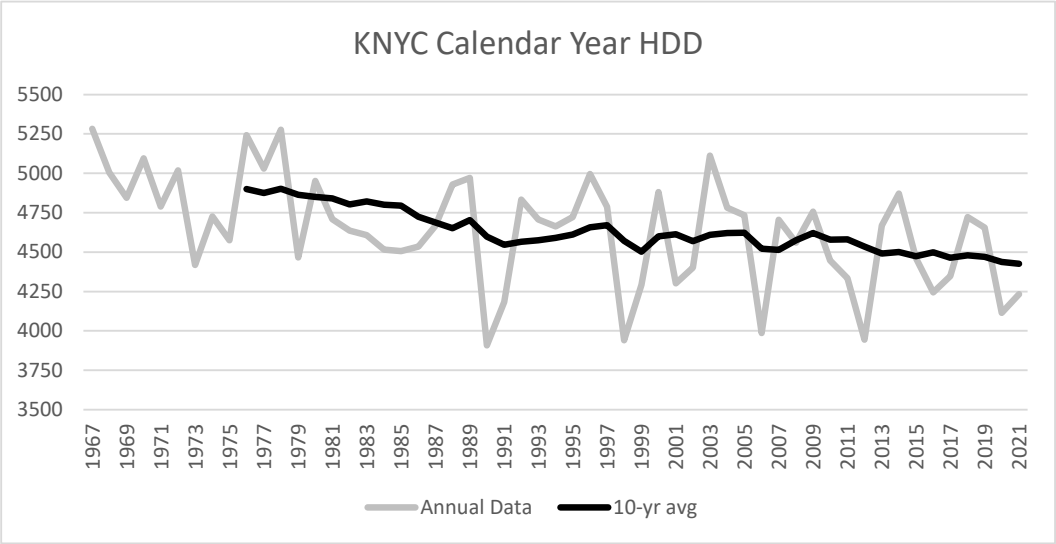


Figure 2, below, shows the calendar year total HDD for the KNYC weather station with a corresponding 10-year normal. The 10-year average in Figure 2 also indicates a slow descent over the time period. However, Figure 2 also shows how

[1] U.S. Climate Normals. National Centers for Environmental Information, NOAA. https://www.ncei.noaa.gov/products/land-based-station/us-climate-normals (last visited September 14, 2023).

the normal can move up and down due to the shorter-term nature of the 10-year normal. Relying on such short-term, annual fluctuations can skew customer costs and not provide the appropriate price signals to consumers.

Figure 2



Q. Do the Companies' agree with the alternate gas load forecasts presented by the SFIP (at 17-23)?

A. No, the Companies do not agree with the SFIP's alternative approach to gas load forecasting.

Q. Please explain why the Companies disagree with the SFIP's alternate version of the gas load forecasts.

A. The Companies believe their forecasts are the best estimation for future meter counts and volumes because the forecasts are based on methodologies that are

rigorous and align with econometric best practices. The Companies' approach involves a range of forecasting methods, specifying appropriate models for each customer group based on several diagnostic checks and accuracy measures. The Companies then refine the models as additional information becomes available each year. The SFIP forecast methodology, on the other hand, is less robust.

Q. Why do the Companies believe the SFIP's alternate forecasting methodology should not be adopted here?

A. There are several reasons why the SFIP's methodology should not be utilized. For example, the SFIP uses the adjusted R-squared to justify its models for forecast Residential Heating meter counts, Multi-Family use-per customer, and Non-Firm Demand Response meter count. However, using the adjusted R-squared is not a good measure of the predictive ability of any model, and should not be used solely as a selection criterion for a forecast model. Adjusted R-squared measures how well the model fits historical data, but not how well the model will forecast future data. When preparing a forecast, it is crucial to measure predictive accuracy of competing models and use the predictive accuracy for selection criteria.

Additionally, the SFIP's approach presents no measures or tests of their models' predictive accuracy in the form of an ex-post analysis. Such an analysis sets aside part of the historical data that forms the test-set and estimates parameters of the

forecasting method using observations from the training set. The training set consists of the historical data that occurred prior to the observations that form the test set and then forecasts for the held-out portion of the historical data. The model will pass the ex-post analysis only if the coefficient estimates from fitting the model on the training set do not vary significantly from the original coefficient estimates that were obtained from fitting the model on the full historical data and based on the forecast accuracy as measured on the test set. As the test set is not used in estimating the model parameters, this ex-post analysis provides a reliable indication of how well the model is likely to forecast on new data. Unlike the SFIP's approach, the Companies' methodology incorporated an ex-post analysis providing assurance to the Commission of the accuracy of their forecasts

Q. Aside from the accuracy concerns with using the adjusted R-squared and not conducting an ex-post analysis, are there other reasons the Companies disagree with the SFIP's alternate forecasting methodology?

A. Yes. The SFIP also does not account for multicollinearity. Multicollinearity occurs when explanatory variables are highly correlated with one another, and it is difficult for the model to distinguish the individual impact of each variable. The Companies' test for multicollinearity because its presence can produce unstable coefficient estimates that are difficult to interpret. For example, potential multicollinearity issues could exist in the SFIP's KEDNY commercial meter

count, where the SFIP uses both the population economic variable and a trend variable.

Q. Do the Companies have any concerns with the SFIP's use of economic variables in its models?

Yes. The SFIP's model for KEDNY residential non-heating meter count transforms Gross Domestic Product ("GDP") and uses it as an explanatory variable. The coefficient on GDP is positive even though there is a negative correlation between GDP and residential non-heating meter counts in the historical data. As GDP increases, the Companies expect more residential customers to install a gas furnace and switch from a non-heating rate to a heating rate. The sign on the coefficient of the SFIP's residential non-heating meter count models contradicts this economic theory and the statistical correlation seen in the data (*See* Exhibit ___ (SFIP-4), at page 5 of 10).

Another example of flawed economic theory is in the SFIP's KEDLI residential heating meter count model (*See* Exhibit ___ (SFIP-5), at page 4 of 10). The coefficient on housing stock in this model is 5,175, which can be interpreted as a one-unit increase (1,000 homes) in housing stock leads to 5,175 more residential heat meters. In theory, the Companies expect some new homes to use delivered

fuels, electric resistance, or heat pumps for heating, meaning that coefficient on housing stock should be less than 1,000.

Q. Do the Companies have any other concerns with the SFIP alternative?

A. Yes. The SFIP's post-model adjustment process was flawed for KEDNY and KEDLI. In KEDNY, the SFIP adopted the volume adjustments for demand-side management and local laws from the Companies' initial filing forecast but did not correctly apply the adjustments to its forecasts. The SFIP incorrectly adjusted Rate Year usage per account forecasts with the fiscal year ("FY") 2024 cumulative adjustments from the Companies' initial filing. The SFIP also only accounted for the volumetric impact of the post-model adjustments and failed to reduce its projected customer counts for meter loss from electrification. In KEDLI, the SFIP used the Companies' corrections and updates forecast assumptions but did not correctly accumulate the meter loss from electrification. This error in KEDLI had a negligible impact on the forecast.

Q. Did the Companies develop a comparison of methodologies after correcting for the errors?

A. Yes. As shown in Tables 1 and 2, below, the Companies corrected the errors in Staff's methodology and produced a new comparison in the Rate Year between the two forecasts.

Table 1: Corrected Comparison between Staff and Companies' Forecast - Rate Year - KEDNY						
KEDNY Volumes (therms) - Rate Year						
	Residential Non-Heat	Residential Heat	Commercial	Multi-Family	NFDR	Total
Staff	55,976,685	764,526,475	298,793,740	395,310,985	105,612,313	1,620,220,198
Company	54,216,365	756,100,775	293,647,325	387,300,392	101,865,678	1,593,130,535
Difference	3.25%	1.11%	1.75%	2.07%	3.68%	1.70%
KEDNY Accounts - Rate Year						
	Residential Non-Heat	Residential Heat	Commercial	Multi-Family	NFDR	Total
Staff	560,737	674,253	53,827	19,797	1,861	1,310,475
Company	552,020	661,400	52,597	19,939	1,771	1,287,727
Difference	1.58%	1.94%	2.34%	-0.71%	5.06%	1.77%

Table 2: Corrected Comparison between Staff and Companies' Forecast - Rate Year - KEDLI						
KEDLI Volumes (therms) - Rate Year						
	Residential Non-Heat	Residential Heat	Commercial	Multi-Family	NFDR	Total
Staff	14,867,496	597,191,355	416,864,680	52,627,859	13,274,616	1,094,826,005
Company	14,770,301	605,451,660	417,611,643	53,107,620	12,793,082	1,103,734,306
Difference	0.66%	-1.36%	-0.18%	-0.90%	3.76%	-0.81%
KEDLI Accounts - Rate Year						
	Residential Non-Heat	Residential Heat	Commercial	Multi-Family	NFDR	Total
Staff	71,040	503,221	61,812	1,774	171	638,019
Company	71,565	501,717	61,784	1,783	170	637,019
Difference	-0.73%	0.30%	0.05%	-0.51%	0.57%	0.16%

Q. Please describe the results after incorporating the corrections.

A. Comparing the corrected models demonstrates that with the proper post-model adjustments, the SFIP's forecast is extremely close to the Companies' (within one percent). The Companies made the following changes to the SFIP's post-model adjustment process:

- Adjusted the SFIP's customer count and use-per-customer forecasts using the post-model adjustments from the Companies' correction and updates

forecast for KEDNY (Exhibit __ (GLF-15ACU) and Exhibit __ (GLF-16ACU)).

- The SFIP compared an annual average of its FY 2025 customer count forecasts to the Companies' March 2025 forecast value for each rate group in KEDNY. The comparison in Tables 1 and 2 above compares FY-end, March 2025, customer counts in both forecasts to create a direct comparison in both KEDNY and KEDLI.
- Used the rate code allocation factors from the Companies' correction and updates forecast in KEDNY. (*See* response to IR No. DPS-1016 provided as Exhibit__(SFIP-1)).
- Correctly accumulated the meter loss from electrification in KEDLI using the assumptions from the Companies' correction and updates forecast in Exhibit __ (GLF-16ACU).

The Companies recognize that no forecasts are perfect and that actuals are likely to be different from any projections. The Companies also acknowledge that the total forecast volume and meters from both parties in the Rate Year are similar (*See* Exhibits___ (SFIP-2) and (SFIP-3)). Nevertheless, the Companies believe their forecasts provide a better indication of future outcomes, because they incorporate a more robust methodology that relies on econometric best practices.

Q. AGREE witness Mark D. Kleinginna's (at 9), proposes modifications to the Companies' design day HDD definition and the forecast KEDNY design day wholesale sendout. Do you agree with his position?

A. No. As stated above, a standard should be a stable reference that does not change significantly over time. The purpose of a design day weather standard is to establish the amount of system-wide throughput (interstate pipeline and underground-storage capacity plus local supplemental capacity) required to maintain the integrity of the distribution system. The testimony and exhibits in the record support the Companies' decision to use 65 HDD as their design day weather standard.

Q. What does Mr. Kleinginna propose regarding reductions to the Companies' forecast design day sendout?

A. Mr. Kleinginna (at 11) proposes three steps to lower the Companies' forecast design day wholesale sendout:

- A reduction using his proposed 30-year occurrence of coldest day;
- A further reduction using his proposed 10-year occurrence of coldest day; and
- A reduction in the rate of growth of the Companies' design day wholesale sendout to match the normal year annual retail volume growth.

Q. Do you agree with the first two points in Mr. Kleinginna's testimony regarding frequency of occurrence of the Companies' design day HDD?

A. No. In proposing that the Companies' design day standard of 65 HDD should be lowered to 57 or 58 HDD, Mr. Kleinginna's analysis ignores statistical probability of occurrence, instead relying on only the most recent coldest day of the most recent winters.

While Mr. Kleinginna does not criticize the Companies' choice of its design day standard, he begins his argument (at 9) stating that the Companies' design day standard is a once-in-89-year occurrence. He derives this rate of occurrence by counting the years since the Companies observed a design day on February 9, 1934. His argument then shifts forward to identify a 61 HDD design day standard as a once-in-30-year occurrence (the dataset used by the Companies for their normal year standard) since a 61 HDD was last observed on January 19, 1994, which was 30 years in the past. Mr. Kleinginna (at 9) then calculates a five-year moving average of annual heating degree days over those 30 years and observes that annual HDD have dropped by 7.7 percent. By suggesting that what happens on the annual level will equally happen on a design day, he reduces his proposed 61 HDD standard down to 57 HDD. Finally, Mr. Kleinginna backpedals to propose a 58 HDD, because a 58 HDD was observed in 2016.

Q. Please explain why the Companies believe a 65 HDD is more appropriate.

A. The Companies chose the 65 HDD design day based on experience and consideration for the important factors that can affect the HDD determination. The Companies' 65 HDD design day was also independently reviewed and supported by Marquette Energy Analytics in June 2021 as part of its Long-Term Capacity Report (Case 19-G-0678). Moreover, the Companies believe the 65 HDD is reasonable, as it sets the point at which service is guaranteed to firm service customers with no portfolio excess. Failure to stand ready for a design day occurrence could have dire consequences of potential curtailments and service outages.

Q. Do you agree with the third point in Mr. Kleinginna's testimony (at 10) regarding the rate of growth of the Companies' design day wholesale sendout?

A. No. Mr. Kleinginna (at 10) observes "that the growth in firm demand for the design day from the 22/23 period through the 27/28 period was 6.94%. However, the growth in retail sales as provided by the gas forecasting panel was 3.04% over FY23 through FY28." He then proposes an additional reduction in KEDNY's design day wholesale sendout to lower its rate of growth to that of overall retail annual volume. This comparison is fundamentally flawed as Mr. Kleinginna is comparing two different datasets. KEDNY's retail volume, to which he refers, is

composed of its Firm retail volumes plus its Non-Firm Demand Response ("NFDR") volumes. From 2022/23 through 2027/28, that total Firm plus NFDR KEDNY volume does grow 3.04 percent per annum (Exhibit___ (GLF-4ACU)). However, as NFDR customers do not flow gas on the design day, KEDNY's design day wholesale sendout volume is only its firm customer load.

From Exhibit___(GLF-4ACU) and Exhibit___(GLF-4BCU), the normal year retail volume growth for KEDNY Firm customers only is 4.81 percent per annum. From AGREE-10, Attachment 1 containing KEDNY's daily design year wholesale sendout requirements, the design year sendout grows at a slightly lower rate of 4.09 percent per annum because the incremental degree days in the design year only affect the KEDNY heating load.

From AGREE-10, Attachment 1, KEDNY's design day sendout does grow at 6.94 percent per annum. The design year Firm (excluding NFDR) wholesale sendout grows at 5.93 percent per annum, in line with the design day growth observed by Mr. Kleinginna. The slightly higher growth rate for the design day is indicative of the Companies' load becoming peakier, either due to incremental loads that are weather sensitive or due to reductions in load in the off-peak periods.

Based on the foregoing, the Panel concludes that all of Mr. Kleinginna's adjustments to KEDNY's design day are flawed and the Companies' design day should be accepted.

Q. Lastly, Mr. Kleinginna's testimony (at 12) discusses the Companies' design day demand forecast in the context of the scenarios presented in the National Grid CLCPA Report. Please discuss the design day as presented in these proceedings with those of the CLCPA Report.

A. The design day wholesale sendout requirements presented in the context of the instant rate filings reflect the current regulatory and environmental status of New York gas planning. As described in the Panel's direct testimony (at 22), upon completion of its initial econometric forecast, "the Companies adjusted the meter count and use-per-customer forecasts for funded energy efficiency programs, heat electrification programs, water heating electrification, as well as the forecast impact of the City of New York Local Laws 154 ("LL154") and 97 ("LL97"), which aim to curtail future gas usage (discussed below in more detail)." The short-term forecast of the Companies' design day sendout requirements reflect the impact of the Companies' currently expected distributed energy resources ("DER") (*e.g.*, energy efficiency programs, electrification of heat, demand response programs) portfolio contribution. The three scenarios presented in the

CLCPA Report were not tied in the short-term to actual funded and/or approved programs.

IV. Conclusion

Q. Does that conclude your rebuttal testimony.

A. Yes.